DEPI

Digital Egypt Pioneers Initiative

Programming Education Platform for Secondary Students

Project Documentation – Part I

Track: Full-Stack .Net Development Technical Instructor: Mohamed Mahmoud
Type: Team Graduation Project
Team Size: 5 Members
Team Leader: Abdelrahman Eslam Ahmed Wajih
GitHub:https://github.com/El7a2ona/LearningPlatform

Submission Schedule (Part I)

Item	DEPI Industry	DEPI
Project Planning & Management	2 Oct 2026	16 Oct 2026
Literature Review	9 Oct 2026	16 Oct 2026
Requirements Gathering	9 Oct 2026	16 Oct 2026

Contents

- 1. Project Planning & Management – Project Proposal, Project Plan, Task Assignment & Roles, Risk Assessment & Mitigation Plan, KPIs
- 2. Literature Review – Review of Similar Solutions, Feedback & Evaluation, Suggested Improvements, Final Grading Criteria
- 3. Requirements Gathering – Stakeholder Analysis, User Stories & Use Cases, Functional Requirements, Non-functional Requirements
- 4. Repository & Project Information

1. Project Planning & Management

1.1 Project Proposal

This project is an online programming education platform for secondary-school students. It provides a structured environment in which students can find suitable programming courses and study content created by teachers. The platform combines course discovery, structured lessons, assessments, progress tracking, feedback, guardian monitoring, and administrative controls for course quality and complaints. The platform also supports the operational needs of teachers and teaching assistants, including course administration, grading, student performance analytics, and the management of eligible earnings from paid courses.

Problem Statement

Learning programming can be challenging for secondary-school students due to the difficulty of finding organized learning resources, following a clear learning path, and evaluating their understanding of programming concepts. Students may also struggle to track their progress, receive timely feedback, and identify areas that require improvement.

At the same time, teachers need an efficient way to organize educational content, manage assessments, grade student submissions, and monitor individual and overall performance. Guardians may also lack a convenient way to follow their children's academic progress, while administrators need mechanisms to maintain course quality and handle complaints.

Therefore, there is a need for an integrated online platform that organizes programming education, simplifies communication between stakeholders, and provides effective tools for learning, assessment, progress tracking, and course management.

Proposed Solution

The proposed solution is to develop a web-based programming education platform that brings students, teachers, teaching assistants, guardians, and administrators together in one integrated system.

The platform will provide the following features:

- Structured learning: Organize courses into modules and lessons, with mandatory quizzes at the end of each module and optional lesson quizzes and assignments.

- Course discovery and enrollment: Allow students to browse courses, view course details and reviews, and enroll in free courses or purchase paid courses.

- Progress tracking and assessment: Record learning activity, quiz attempts, assignment submissions, grades, and feedback through student dashboards.

- Course management: Enable teachers to create and manage courses, prepare learning materials, assign teaching assistants, and evaluate student performance.

- Teaching assistant support: Provide access to assigned courses and tools for grading authorized student submissions.

- Guardian monitoring: Allow guardians to monitor their verified children's enrolled courses, progress, and grades.

- Administrative control: Support paid-course approval, complaint management, review moderation, and refund-request handling.

- Payment and financial management: Track eligible teacher earnings, apply refund periods, and manage payout holds according to course policies.

Through these features, the platform will provide a centralized environment for delivering, managing, and monitoring programming education

Objectives

The main objectives of the project are:

1. To provide secondary-school students with an organized and accessible platform for learning programming.
2. To simplify the process of discovering, selecting, and enrolling in suitable programming courses.
3. To support structured learning through modules, lessons, quizzes, and assignments.
4. To monitor students' academic progress and provide clear access to grades and feedback.
5. To help teachers create educational content, manage courses, and evaluate student performance efficiently.
6. To enable teaching assistants to support assessment and grading within their assigned courses.
7. To allow guardians to monitor their authorized children's academic progress.
8. To maintain educational quality through administrative course approval, complaint handling, and review moderation.
9. To support secure enrollment and payment processes for both free and paid courses.
10. To protect users' academic and financial information through role-based access control and reliable data management.
11. To provide dashboards and analytics that help teachers understand student and course performance

Scope

- **In scope:** user registration and login; course browsing, course details, and course structure; free and paid courses, enrollment, and payment; course modules, lessons, mandatory module quizzes, optional lesson

quizzes, and lesson assignments; student progress tracking, assignment submission, quiz attempts, grading, grades, and feedback; course reviews and ratings; teacher course management and teacher dashboards; teaching assistant assignment, assigned-course access, and grading; guardian monitoring of linked children; student complaints and administrative complaint handling; administrative approval of paid courses and teacher compliance acknowledgement; refund periods, delayed teacher payout, and teacher balance; course lifecycle management.

- **Optional (MVP-dependent):**
  Out of scope: live code editor, programming assignments (programming problems), and general programming tracks beyond the school curriculum. These are planned for future versions and are outside the current core scope unless separately approved as a change to the SRS..

Technology Stack

Area	Technology
Frontend	HTML, CSS, JS
Backend	ASP.NET
Database	SQL Server
Authentication	JWT
Version Control	Git + GitHub, feature branches + Pull Requests

1.2 Project Plan

The project runs for **9 weeks, from 4 Oct to 4 Dec 2026**. The plan covers the whole project (analysis, design, implementation, testing, and final delivery) and is independent of the individual documentation deadlines. Implementation is finished and deployed by **28 Nov**, ahead of the 30 Nov implementation deadline.

Phases

Phase	Dates	Goal
1. Planning & Setup	4 – 17 Oct (W1–W2)	Idea, scope, repository, branches, task split, data models and API contracts agreed
2. Backend Foundation	18 – 24 Oct (W3)	Authentication, core models, and core APIs working and tested in Postman
3. Core Features	25 Oct – 7 Nov (W4–W5)	Main screens and APIs built for each feature, including AI analysis and chat
4. Integration & Completion	8 – 21 Nov (W6–W7)	Frontend connected to backend, all features complete, feature freeze
5. Testing & Deployment	22 – 28 Nov (W8)	Test cases, bug fixing, deployment, README
6. Final Delivery	29 Nov – 4 Dec (W9)	Final reports, presentation, demo, rehearsal

Gantt Chart

Workstream	W1	W2	W3	W4	W5	W6	W7	W8	W9
Planning, SRS and repo setup									
Auth and role-based access									
Courses, modules and lessons									
Enrollment, access rules and student progress									
Quizzes, assignments and grading (including teaching assistants)									
Reviews, complaints and admin moderation									
Dashboards (teacher, teaching assistant, guardian)									
Paid courses, payments, refunds and payouts									
Shared UI and integration									
Testing and bug fixing									
Deployment and README									
Final reports and presentation									
Buffer & Rehearsal									

W1 = 4–10 Oct, W2 = 11–17 Oct, W3 = 18–24 Oct, W4 = 25–31 Oct, W5 = 1–7 Nov, W6 = 8–14 Nov, W7 = 15–21 Nov, W8 = 22–28 Nov, W9 = 29 Nov–4 Dec.

Milestones

#	Milestone	Date
M1	SRS baseline, repo and API contracts agreed	17 Oct
M2	Auth with roles, plus User, Course, Module and Lesson models	24 Oct
M3	Course browsing, free enrollment and teacher course creation	31 Oct
M4	Quizzes, assignments, submission and grading (teacher and teaching assistant)	7 Nov
M5	All five features integrated	14 Nov
M6	Feature freeze (all features complete)	21 Nov
M7	Code freeze and deployed version live	28 Nov
M8	Final presentation, demo, and reports ready	4 Dec

Deliverables

Deliverable	Owner	Due
GitHub repository with branches and PR history		Continuous
Backend APIs for all five features		21 Nov

Deliverable	Owner	Due
Frontend for all screens		21 Nov
Admin dashboard		21 Nov
Test cases, test plan, bug reports		28 Nov
API documentation		28 Nov
Deployed web app and README		28 Nov
Technical Documentation		3 Dec
User Manual		3 Dec

Deliverable	Owner	Due
Project Infographic		3 Dec
Presentation (PPT/PDF)		3 Dec
Video demo (optional)		3 Dec

Resource Allocation

Resource	Allocation
Team	5 developers; each owns one feature across frontend, backend, database, API, and integration
Version control	GitHub with feature branches. No direct push to main; every merge goes through a reviewed Pull Request
External services	SQL Server
Meetings	Weekly planning (Sunday), mid-week sync on blockers, Thursday demo and merge of reviewed PRs
Review process	At least one other developer reviews each Pull Request before merging

1.3 Task Assignment & Roles

Team Leader: Gamila Abdel Khalek. Each developer owns a complete feature from frontend to database and API integration, so nobody is assigned to frontend or backend only.

DEV	Member	Area	Main Responsibilities
DEV 1	Abdelrahman Eslam		
DEV 2	Tameer Saeed		
DEV 3	Malak Abdulhamid		

DEV	Member	Area	Main Responsibilities
DEV 4	Mohamed Ahmed		
DEV 5	Ola Sherif		

1.4 Risk Assessment & Mitigation Plan

Risk	Impact	Likelihood	Mitigation
Integration conflicts between features	High	Medium	Agree on API contracts early; use feature branches and Pull Requests; integration weeks W4–W7
Limited experience with some backend concepts	Medium	Medium	Work incrementally, review code together, test each feature before integration
Scope becoming too large	High	Medium	Prioritize the MVP; cut Exchange Proposal first if needed (see Scope Control)
Merge or regression issues	Medium	Medium	Require self-testing and code review before merging

Risk	Impact	Likelihood	Mitigation
Time pressure near deadlines	High	High	Track milestones weekly; feature freeze 21 Nov; deployment by 28 Nov; buffer week W9
A team member unavailable or behind schedule	High	Low	Raise delays immediately; rebalance tasks or reduce scope; shared ownership of reviews
Deployment or hosting problems	Medium	Medium	Start deployment preparation in W7; keep a local run guide in the README

1.5 Key Performance Indicators (KPIs)

KPI	Target	How It Is Measured
API response time	Under 2 seconds for browsing and search courses.	Postman / browser network tab
Search and filter accuracy	Results match the selected filters and show only active, approved listings	Test cases
Authentication and authorization	Protected routes reject unauthenticated users; admin routes reject non-admins	Test cases
Test coverage	Test cases written and run for every feature; critical bugs fixed before release	Test plan and bug report
System availability	Deployed app accessible during the evaluation period	Manual checks, deployment link
Repository and documentation	All source code and documents in GitHub; every merge via reviewed PR	Repository history
User adoption (pilot)	Pilot test with students and teachers; positive feedback on core flows	Short feedback form after deployment

2. Literature Review

2.1 Feedback & Evaluation

This section records the lecturer's assessment of the project. It is completed after the project is presented to the Technical Instructor, Mohamed Mahmoud.

Area	Lecturer Feedback	Team Response
Project idea and scope		
Planning and documentation		
Requirements		
Overall evaluation		

2.2 Suggested Improvements

Areas where the project can be enhanced, based on the team's own review (to be extended with the lecturer's suggestions):

- Add a mobile application.
- Add AI assistant
- Live Code Editor
- Programming Assignments
- General Programming Tracks

2.3 Final Grading Criteria

The official breakdown of marks across documentation, implementation, testing, and presentation is provided by the Technical Instructor. The table is completed once the grading scheme is confirmed.

Component	Marks	Related Project Deliverables
Documentation		Planning, literature review, requirements, system analysis and design, reports
Implementation		Source code, GitHub repository, deployed application, README
Testing		Test cases and test plan, automated tests, bug reports
Presentation		Presentation (PPT/PDF), infographic, optional video demo
Total		

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs	Influence / Interest
Secondary-school students	Discover and study programming courses; enroll or purchase; submit quizzes and assignments; view grades and progress; review courses; raise complaints.	High Interest, Medium Influence
Teachers	Create and manage courses and learning materials; create assessments; assign teaching assistants; grade work; monitor performance; view eligible earnings.	High Interest, High Influence
Teaching Assistant	Support assigned courses by viewing students, monitoring performance, and grading authorized assignments and quizzes.	High Interest, Medium Influence
Guardians	Monitor the enrolled courses, progress, quiz grades, and assignment grades of linked and verified children.	High Interest, Medium Influence
Administrators	Review paid courses, manage complaints and review moderation, review refund requests, and enforce platform policies.	High Interest, High Influence
Payment provider	Support course payment and refund transactions and provide transaction status to the platform.	Medium Interest, High Influence

3.2 User Stories & Use Cases User Stories

- As a student, I want to register and log in so that I can securely use my account.
- As a student, I want to browse active listings so that I can discover available items.
- As a student, I want to search and filter listings so that I can quickly find relevant products.
- As a student, I want to save listings as favorites so that I can return to them later.
- As a student, I want to create and manage my listings so that I can sell items.
- As a student, I want AI assistance when creating a listing so that product information can be filled more efficiently.
- As a student, I want to mark my item as sold so that other students know it is no longer available.
- As a student, I want to create a request for an item I need so that other students can respond.
- As a student, I want to communicate with another student so that I can discuss a listing or request.
- As an administrator, I want to moderate listings so that inappropriate or invalid content can be controlled.
- As an administrator, I want to manage users and categories so that the platform remains organized.
- As an administrator, I want to review reports so that I can act on reported content.

Use Cases

Actor	Use Case	Description
Student	Register / Login	Create an account and authenticate
Student	Browse Listings	View available marketplace items
Student	Search & Filter	Find listings using search and marketplace filters
Student	Manage Favorites	Save or remove listings from favorites
Student	Manage Listings	Create, edit, view, and delete owned listings; mark as sold
Student	Create Request	Publish a request for an item needed
Student	Chat	Communicate with other students
Student	Propose Exchange (optional)	Propose an exchange; the other student accepts or declines
Admin	Manage Users	Search, filter, block, and unblock users
Admin	Moderate Listings	Approve or reject listings
Admin	Manage Reports	Review and manage reported content
Admin	Manage Categories	Create, update, and delete categories

3.3 Functional Requirements

FR-AUTH-01: The system shall allow users to log in using a username and password.

FR-AUTH-02: The system shall allow users without an account to create a new account.

FR-AUTH-03: During registration, the user shall select their role. The currently supported public roles are: Student; Teacher.

FR-AUTH-04: The system shall collect different information depending on the selected role.

FR-AUTH-05: The system shall authenticate users before allowing access to protected functionality.

FR-AUTH-06: Users shall only have access to functionality corresponding to their role.

FR-AUTH-07: Administrator account creation shall be controlled by the platform and shall not be treated as normal public registration.

FR-COURSE-01: The system shall display available courses using course cards. A course card may contain: Course image; Course name; Teacher name; School grade; Term; Course price; Free/Paid status.

FR-COURSE-02: Students shall be able to browse courses without being forced to select grade or term before browsing.

FR-COURSE-03: The system shall provide a course details page.

FR-COURSE-04: The course details page shall display relevant information such as: Course name; Teacher; Grade; Term; Course description; Number of enrolled students; Modules; Course ratings; Course reviews; Course price/status.

FR-COURSE-05: Students shall be able to inspect the course structure before enrollment.

FR-COURSE-06: A course shall contain grade and term information when it represents a school curriculum.

FR-COURSE-07: A course may cover: The complete curriculum; A part of the curriculum; A specific topic or group of topics.

FR-COURSE-08: Teachers shall be free to organize modules according to their teaching methodology. Modules do not have to exactly match the official curriculum units.

FR-CONTENT-01: A course shall contain one or more modules.

FR-CONTENT-02: Each module shall contain a mandatory module-level quiz.

FR-CONTENT-03: Each lesson shall support an assignment. The teacher decides whether to provide an assignment for the lesson.

FR-CONTENT-04: The teacher may optionally add a quiz to an individual lesson.

FR-CONTENT-05: Students shall be able to view module names and the available lesson/assessment structure before enrollment.

FR-CONTENT-06: Teachers shall be able to create and manage modules.

FR-CONTENT-07: Teachers shall be able to create and manage lessons.

FR-CONTENT-08: Teachers shall be able to provide external lecture/video. The platform does not need to host uploaded video files.

FR-CONTENT-09: Teachers shall be able to create assignments for lessons.

FR-CONTENT-10: Teachers shall be able to create mandatory module quizzes.

FR-CONTENT-11: Teachers shall be able to create optional lesson quizzes.

FR-ENROLL-01: A student shall be able to enroll in multiple courses.

FR-ENROLL-02: The system shall clearly indicate whether a course is: Free; Paid.

FR-ENROLL-03: When a student enrolls in a free course, the student shall receive immediate access to the course content.

FR-ENROLL-04: When a student successfully purchases a paid course, the student shall receive immediate access to the course content.

FR-ENROLL-05: The student's enrolled courses shall be available from the student's account.

FR-ACCESS-01: The student shall be able to start learning from any module. The student does not have to complete Module 1 before accessing Module 2.

FR-ACCESS-02: Lessons inside the selected module shall follow sequential access.

FR-ACCESS-03: The student shall not be able to open the next lesson until the content of the previous lesson has been completed.

FR-ACCESS-04: Watching a lesson shall not necessarily be required before accessing its quiz or assignment. Therefore, the student may access a quiz or assignment without watching the corresponding lecture when the system's access rules allow it.

FR-ACCESS-05: The student shall not be required to wait for an assignment to be graded before continuing to the next available learning step.

FR-PROGRESS-01: The system shall track the student's progress within each enrolled course.

FR-PROGRESS-02: Progress shall take into consideration assessment activity, including: Quizzes; Assignments.

FR-PROGRESS-03: Video/lesson viewing activity shall also be tracked where applicable.

FR-PROGRESS-04: The system shall store student assessment results.

FR-PROGRESS-05: Students shall be able to view their own progress.

FR-PROGRESS-06: Teachers shall be able to view student progress.

FR-PROGRESS-07: Teaching assistants shall be able to view student progress for courses they assist in.

FR-PROGRESS-08: Guardians shall be able to view their child's progress.

FR-PROGRESS-09: The exact mathematical formula used to calculate the overall progress percentage shall be defined before implementation.

FR-QUIZ-01: Each module shall contain a mandatory quiz.

FR-QUIZ-02: Teachers may optionally add quizzes to individual lessons.

FR-QUIZ-03: Students shall be able to attempt available quizzes.

FR-QUIZ-04: The system shall store quiz results.

FR-QUIZ-05: Teachers shall be able to view student quiz results.

FR-QUIZ-06: Teaching assistants shall be able to view and grade authorized quizzes.

FR-QUIZ-07: Quiz results shall contribute to the student's course performance.

FR-ASSIGN-01: Teachers shall be able to add an assignment to a lesson.

FR-ASSIGN-02: Assignments are optional for the teacher.

FR-ASSIGN-03: Students shall be able to submit assignments.

FR-ASSIGN-04: After submitting an assignment, the student shall be allowed to continue to the next available step without waiting for grading.

FR-ASSIGN-05: Teachers shall be able to view submitted assignments.

FR-ASSIGN-06: Teaching assistants shall be able to view submitted assignments for their assigned courses.

FR-ASSIGN-07: Teachers and authorized teaching assistants shall be able to grade assignments.

FR-ASSIGN-08: The system shall store assignment grades.

FR-ASSIGN-09: Students shall be able to view their assignment grades after grading.

FR-REVIEW-01: The system shall support course ratings and written reviews.

FR-REVIEW-02: A student shall only be allowed to review a course after completing at least one module.

FR-REVIEW-03: Course ratings and reviews shall be displayed on the course details page.

FR-REVIEW-04: The exact review moderation policy shall be defined before implementation.

FR-TEACH-01: A teacher shall be able to create a course.

FR-TEACH-02: A teacher shall be able to create modules.

FR-TEACH-03: A teacher shall be able to create lessons.

FR-TEACH-04: A teacher shall be able to add lecture/video.

FR-TEACH-05: A teacher shall be able to create quizzes.

FR-TEACH-06: A teacher shall be able to create assignments.

FR-TEACH-07: A teacher shall be able to manage their courses.

FR-TEACH-08: A teacher shall be able to add teaching assistants to a course.

FR-DASH-01: The teacher dashboard shall display student/course analytics.

FR-DASH-02: Analytics shall be presented using charts where appropriate.

FR-DASH-03: The teacher shall be able to view all students enrolled in their courses.

FR-DASH-04: The teacher shall be able to select a specific student.

FR-DASH-05: For a selected student, the teacher shall be able to view: Quiz grades; Assignment grades; Course progress; Learning activity; Number of videos/lessons viewed where tracked.

FR-DASH-06: The teacher shall be able to view course-level student performance.

FR-DASH-07: The dashboard shall display the teacher's balance.

FR-DASH-08: The teacher balance shall represent money accumulated from eligible course purchases according to the platform's payout policy.

FR-TA-01: A teacher shall be able to add a teaching assistant to a course.

FR-TA-02: The system shall grant the teaching assistant access to the assigned course.

FR-TA-03: The teaching assistant shall see all courses in which they are assigned.

FR-TA-04: The teaching assistant shall be able to open an assigned course.

FR-TA-05: The teaching assistant shall be able to view all students in the course.

FR-TA-06: The teaching assistant shall be able to view assignments that have not been graded.

FR-TA-07: The teaching assistant shall be able to view quizzes requiring grading.

FR-TA-08: The teaching assistant shall be able to grade assignments.

FR-TA-09: The teaching assistant shall be able to grade quizzes where applicable.

FR-TA-10: The teaching assistant shall be able to view student grades and course performance.

FR-TA-11: The teaching assistant shall not have unrestricted administrative permissions.

FR-GUARD-01: The guardian shall be able to view all courses in which their child is enrolled.

FR-GUARD-02: The guardian shall be able to view the child's progress in each course.

FR-GUARD-03: The guardian shall be able to view the child's quiz grades.

FR-GUARD-04: The guardian shall be able to view the child's assignment grades.

FR-GUARD-05: The exact mechanism used to link a guardian account to a student's account shall be defined before implementation.

FR-COMPLAINT-01: The student shall be able to submit a complaint or feedback regarding a teacher/course.

FR-COMPLAINT-02: The administrator shall receive the complaint.

FR-COMPLAINT-03: The administrator shall review the complaint.

FR-COMPLAINT-04: The administrator shall be able to contact the relevant teacher.

FR-COMPLAINT-05: The administrator shall investigate and handle the complaint according to platform policies.

FR-COMPLAINT-06: The administrator shall record the complaint status and outcome.

FR-PAID-01: Teachers shall be able to create paid courses.

FR-PAID-02: A paid course shall not become available for purchase before administrative approval.

FR-PAID-03: The administrator shall contact the teacher during the approval process.

FR-PAID-04: The administrator shall verify that the teacher is serious about providing the course.

FR-PAID-05: The administrator shall verify that at least the first module is prepared.

FR-PAID-06: The teacher shall receive an official acknowledgement/undertaking before approval.

FR-PAID-07: The acknowledgement shall state that the teacher is required to follow: The declared course content; Applicable company policies.

FR-PAID-08: The acknowledgement shall state that violating the agreed content/policies may result in a financial penalty according to company policy.

FR-PAID-09: After successful approval, the paid course shall become available for students to purchase.

FR-PAY-01: Students shall receive immediate access after successfully purchasing a paid course.

FR-PAY-02: The teacher shall not immediately receive the corresponding funds in their available balance.

FR-PAY-03: For courses where content is released progressively, the payout shall remain on hold for 7 days

after the student's purchase.

FR-PAY-04: The seven-day period represents the period during which the student has the right to request a refund.

FR-PAY-05: For fully recorded courses, the refund/payout hold period shall be 24 hours.

FR-PAY-06: After the applicable refund period passes, the corresponding amount becomes available to the teacher according to the platform's payout policy.

FR-PAY-07: The teacher's balance shall represent funds that have become available according to the payout rules.

FR-PAY-08: The exact payment provider and financial transaction mechanism shall be defined separately.

FR-LIFECYCLE-01: A course shall have a lifecycle of one year.

FR-LIFECYCLE-02: After the one-year lifecycle ends, the course shall be deleted according to the platform's course lifecycle policy.

FR-LIFECYCLE-03: The exact starting point used to calculate the one-year period shall be finalized before implementation.

FR-LIFECYCLE-04: The system shall define how historical student grades and financial records are handled when a course reaches the end of its lifecycle.

BR-01: A student can enroll in multiple courses.

BR-02: Free courses provide immediate access after enrollment.

BR-03: Paid courses provide immediate access after successful payment.

BR-04: Students can start from any module.

BR-05: Lessons inside a module are sequential.

BR-06: The student does not have to watch a lesson before accessing its quiz or assignment when access is available.

BR-07: Every module has a mandatory quiz.

BR-08: Lesson-level quizzes are optional.

BR-09: Lesson assignments are optional for the teacher.

BR-10: The student does not have to wait for assignment grading before continuing.

BR-11: A student must complete at least one module before reviewing a course.

BR-12: Teaching assistants can grade assignments and quizzes for their assigned courses.

BR-13: Paid courses require administrative approval before publication.

BR-14: Teacher payout is delayed to protect the student's refund period.

BR-15: The payout hold is 7 days for progressively released courses.

BR-16: The payout hold is 24 hours for fully recorded courses.

BR-17: A course has a one-year lifecycle.

3.4 Non-functional Requirements

Category	Requirement
Security	NFR-SEC-01: Passwords shall never be stored as plaintext. NFR-SEC-02: The system shall use secure authentication mechanisms. NFR-SEC-03: Role-based authorization shall prevent unauthorized access. NFR-SEC-04: Students shall only access their own private academic information. NFR-SEC-05: Teachers shall only manage courses and students associated with their courses. NFR-SEC-06: Teaching assistants shall only access courses to which they are assigned. NFR-SEC-07: Guardian access shall be limited to authorized children. NFR-SEC-08: Administrative functions shall only be accessible to authorized administrators.
Performance	NFR-PERF-01: The system should provide responsive navigation during normal usage. NFR-PERF-02: Course browsing should remain responsive as the number of courses increases. NFR-PERF-03: Teacher dashboards and analytics should load within an acceptable time. NFR-PERF-04: Assignment and quiz submissions shall provide clear confirmation of successful submission.
Usability	NFR-UX-01: The user interface shall clearly distinguish between different user roles. NFR-UX-02: Course cards shall provide important course information at a glance. NFR-UX-03: Students shall easily understand: Current module; Current lesson; Available next lessons; Pending assignments; Available quizzes; Progress. NFR-UX-04: Teacher dashboards shall clearly display important analytics. NFR-UX-05: Teaching assistants shall easily identify ungraded assignments and quizzes. NFR-UX-06: Error messages shall be clear and understandable to non-technical users.
Reliability	NFR-REL-01: The system shall minimize data loss. NFR-REL-02: Quiz submissions shall not be silently lost. NFR-REL-03: Assignment submissions shall not be silently lost. NFR-REL-04: Grades shall be stored reliably. NFR-REL-05: Financial transactions shall be traceable. NFR-REL-06: Important administrative actions should be auditable.
Maintainability	NFR-MAINT-01: The system should be designed so future educational features can be added without major redesign. NFR-MAINT-02: The system should support future programming tracks. NFR-MAINT-03: The system should allow future integration of the live code editor. NFR-MAINT-04: Business rules such as payout periods and course policies should be maintainable as platform policies evolve.
Scalability	NFR-SCALE-01: The system should support increasing numbers of: Students; Teachers; Courses; Teaching assistants; Assessments; Enrollments. NFR-SCALE-02: The increase in users and course activity should not cause unacceptable degradation of normal system functionality.
Compatibility	
Localization	
Privacy	NFR-PRIV-01: Student academic information shall only be accessible to authorized users. NFR-PRIV-02: Guardian information shall only expose data belonging to the linked child. NFR-PRIV-03: Private student information shall not be publicly displayed on course pages. NFR-PRIV-04: The system shall have defined policies for data retention and deletion.

Category	Requirement
Financial Integrity	NFR-FIN-01: Payment records shall be accurate. NFR-FIN-02: Teacher balances shall be calculated from valid transactions. NFR-FIN-03: Refund-related transactions shall be traceable. NFR-FIN-04: Payout-hold periods shall be applied consistently. NFR-FIN-05: The system shall prevent accidental duplication of financial transactions.

4. Repository & Project Information

GitHub Repository: https://github.com/El7a2ona/LearningPlatform

The repository contains the project source code and project documentation. The team maintains a feature-branch workflow and uses Pull Requests for controlled integration.

Next Documentation Phase

The next phase expands the requirements into System Analysis & Design deliverables: software architecture, ER diagram and schema, use case diagram, DFD, sequence, activity, state and class diagrams, UI/UX mockups, component and deployment views, API documentation, and the testing plan.